Classification of left and right hand motor imagery tasks based on EEG frequency component selection

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***Abstract*—in this paper, a method based on the time-frequency analysis of EEG frequency spectral Fisher-ratio is proposed to pre-select the most relevant movement-related EEG features. Within this method, combining EEG spectral time-frequency distribution with Fisher criterion, the detailed separability information of the frequency components between two classes of EEG patterns in time-frequency plane over C3, C4, Cz are well characterized, which provides a good guide for selecting the most relevant EEG frequency components. According to Fisher-ratio distribution of EEG spectrum by Matching Pursuit (MP) with high frequency resolution, the matched method Morlet wavelet filter is applied to extract the most relevant EEG frequency components. With the optimized EEG features, two classes of EEG patterns during left and right hand motor imagery are discriminated. Here, BCI competition data are analyzed offline and the satisfactory classification results are obtained, which verify the effectiveness of the proposed method in selecting the most relevant EEG spectral components.**

Keywords-Fisher-ratio; Matching Pursuit; EEG optimization; Morlet wavelet; BCI;

I. INTRODUCTION

Brain Computer Interface (BCI) provides a new communication and control channel between the brain and external environment which does not depend on the brain's normal output pathways of peripheral nerves and muscles so that it can offer the effective help for those with motor disabilities [1]. Studies have shown that the unilateral hand motor imagery will result in ERD/ERS (event-related desynchronization /synchronization) over contralateral and ipsilateral sensorimotor cortex respectively [2]. Based on the antagonistic ERD/ERS patterns, the left and right hand motor imagery tasks could be identified. The movement-related EEG rhythms, especially μ and central β rhythms are associated with those cortical areas most directly connected to the brain's normal motor output channels, which can be used to support an EEG-based BCI system (BCIs) [3].

A BCIs with satisfactory classification accuracy depends not only on good performance of the classifier but also on the choice of the proper EEG features. Therefore, the choice of suitable and reliable features of the available EEG signals is crucial for good BCI communication. Pregenzer et al showed that the classification results depend strongly on an appropriate preselection of EEG spectral components [4]. In this study, MP (Matching Pursuit) with higher frequency resolution combined with Fisher criterion are applied to get the intuitionistic separability of EEG spectral components corresponding to two classes of EEG patterns changing with time. Then combining with the matched Morlet wavelet for extracting the most relevant EEG spectral components, the left and right hand motor imagery tasks are discriminated and the satisfactory classification results are obtained, which verifies the effectiveness of the proposed method.

II. EXPEREMENT DATA

The experimental data are from BCI2003 competition website provided by Graz University of technology. The experiment process is as follows: The event-related EEG changes were investigated in a feedback-guided motor imagery experiment. The subject performed the following task repeatedly in a series of sessions, in which each trial lasted 9 sec. During the first 3 sec reference period of each trial, the subject was asked to keep relaxed with eyes open, followed by an arrow pointing either to the right or left (cue stimulus) indicating motor imagery task of either with right or left hand till t=4.25s (t stands for time). The feedback bar, presented during the following period till t=9s, was moving horizontally towards the right or left boundary of the screen dependent on the on-line classification of the EEG signals, which directs the subject performing hand motor imagery. The feedback is the classification result obtained by the analysis of the preceding 1s EEG. Three bipolar EEG channels were measured over the anterior and posterior of C3, Cz, C4 with inter-electrode intervals of 2.5cm. EEG was sampled with 128Hz and filtered between 0.5 and 30Hz. The dataset were obtained from BCI2003 competition website provided by Graz University of technology. EEG was collected from a normal subject, which is denoted as Sub_Graz. The MATLAB files, x_train.mat, y_train.mat are train datasets and x_test.mat, y_test.mat are test datasets, which both include 140 trials and the class label respectively. The details could be found in [5].

III. METHODS

A. Fisher criterion for class separability measure

To search for EEG spectral components most related to motor imagery, Fisher criterion is adopted. The expected EEG spectral components should be those that have best separability between left and right hand motor imagery tasks.

Fisher-ratio as an effective class separability measure has been widely used, in which two measures related to class separation include the intra-class difference and interclass

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spread. Let the mean and variance of samples belonging to class C_1 and class C_2 in the direction of the ith feature be denoted by m_{1i}, m_{2i} , σ_{1i}, σ_{2i} , respectively. Fisher-ratio between class C_1 and C_2 for the ith feature is defined as the ratio of the interclass difference to the intra-class spread.

$$fr_i = \frac{(m_{1.i} - m_{2.i})^2}{(\sigma_{1.i}^2 + \sigma_{2.i}^2)} \tag{1}$$

where fr_i is Fisher-ratio and denotes the class separation between classes C_1 and C_2 in the direction of the ith feature. Fisher-ratio provides a good class separability measure because it is maximized with the interclass difference being maximized and the intraclass spread being minimized [6].

Before applying Fisher criterion, the candidate feature sets must be specified. In previous studies [2-5], it was shown that μ and central β rhythms are closely related to the movement-related EEG. Therefore the future analysis are restricted to frequency components from 0 to 30Hz.

B. Fisher-Ratio time-frequency mapping based on MP algorithm

MP is a powerful tool for investigating the non-stationary signal, which was proposed by Mallat et al. [7]. MP can provide high-resolution energy time-frequency distribution of a signal by selecting time-frequency atoms from a dictionary of functions, generally Gabor time-frequency atoms, which best match with the signal structures. Here, to reveal finely the separability of EEG spectral components between two classes of EEG patterns, MP is used to get the time-frequency energy distribution, and further combining with Fisher criterion, MP-Fisher-ratio (MFR) time-frequency distributive mapping is obtained, which can be described as follows:

1. The movement-related EEG signal of each trial during 9s period is processed by MP algorithm to obtain the corresponding time-frequency energy distribution of EEG signal $\mathbf{F}_{tf}(t, f)$.
2. According to the class label, the mean and standard deviation of each element in two classes of samples $\mathbf{F}_{tf}(t, f)$ are calculated respectively to obtain the corresponding mean matrix $\mathbf{M}_{tf(1)}(t, f), \mathbf{M}_{tf(2)}(t, f)$ and deviation matrix $\mathbf{S}_{tf(1)}, \mathbf{S}_{tf(2)}$;
3. By Equation (1), MP-Fisher-Ratio i.e. the matrix **MFR** is computed as follows:

$$\mathbf{MFR}(t, f) = \frac{[\mathbf{M}_{tf(1)}(t, f) - \mathbf{M}_{tf(2)}(t, f)]^2}{\mathbf{S}_{tf(1)}(t, f) + \mathbf{S}_{tf(2)}(t, f)} \tag{2}$$

where $\mathbf{MFR}(t, f)$ reflects the separability of EEG frequency component between two classes of EEG patterns at t time point and frequency f . Here, the frequency resolution is selected as 0.125Hz.

C. Feature extraction and classification

MFR time-frequency mapping can display intuitively the distribution of separability degree of EEG spectral components between two classes of EEG patterns so that it provides the useful information for guiding pre-selection of EEG spectral components mostly related to hand motor imagery. How to express the important information and how to extract these relevant EEG features accurately are key to the further classification. This requires that the feature extraction method matches the process of getting MFR time-frequency mapping.

In MP algorithm, the time-frequency energy distribution of a signal can be represented as a linear combination of basic waveforms that are selected from a dictionary of Gabor functions. These waveforms best match the signal structures so that the information reflected in MFR time-frequency mapping is closely related to the Gabor time-frequency atom. Morlet wavelet takes a similar function form with Gabor time-frequency atom, which is well localized both in time and frequency. Therefore, Morlet wavelet method is used as a filter to extract the relevant EEG spectral features.

Morlet wavelet function takes the form as follows [8]:

$$\psi_{\tau,s,f_0}(t) = \frac{1}{\sqrt{s}} \pi^{-\frac{1}{4}} e^{j2\pi f_0 t} e^{-\frac{1}{2}(\frac{t-\tau}{s})^2} \tag{3}$$

where τ, s stand for the time center and scale factor which control time point and frequency width. The spectrum of Morlet wavelet function is a Gauss function with the center frequency f_0 . By modulating the parameter f_0 , the local characteristic near the frequency f_0 of the signal $G(t)$ to be filtered could be flexibly adjusted. Two important parameters including frequency f_0 and scale factor s respectively determine the center frequency of the signal to be filtered and the frequency bandwidth of a filter. By computing convolution with original EEG signal, EEG spectral components within the specific frequency band can be obtained [8]:

$$F(\tau, f) = \frac{1}{\sqrt{s}} \left\| G(t) * \psi_{\tau,s,f_0}(t) \right\| \tag{4}$$

where $F(\tau, f)$ is the time varying EEG spectrum component of EEG signal $G(t)$.

According to MFR time-frequency mapping, using Morlet wavelet, the optimized EEG feature vector could be constructed as follows. Firstly according to MFR time-frequency mapping over C3, C4, Cz, the several relevant narrow frequency bands corresponding to large Fisher-ratio values are specified; then based on the relevant frequency bandwidth, the center point of the frequency band is specified as the characteristic frequency parameter f_0 of Morlet wavelet. The spread degree of the relevant frequency range relative to the center frequency point determines the selection of scale factor s . The wider of the frequency bandwidth spread, the smaller of the scale factor s and vice visa. For

EEG over different electrodes, the parameters f_0 and S are selected individually. With the specified parameters, EEG signals are filtered with the specific Morlet wavelet function to obtain the optimized frequency components features. Finally combining these EEG features within different narrow frequency bands over three electrodes, the multidimensional optimized EEG feature vector is constructed.

D. *Classification of left and right hand motor imagery taskss*

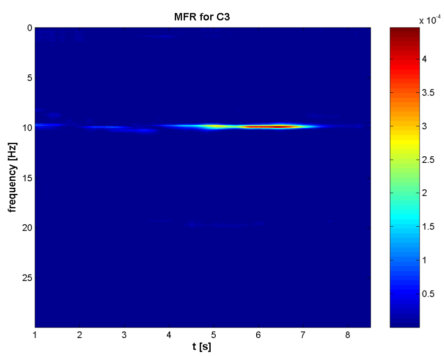
Here, Fisher Discriminant Analysis (FDA) is used to realize identification of two classes of EEG patterns, in which the discriminant distance across all the preceding time are incorporated, leading to an information accumulation for improving classification performance. The details can be found in [9].

The two major indexes i.e. classification accuracy and Mutual Information (MI) are used for not only evaluating classification performance but also reflecting the separability of features between two classes of EEG patterns.

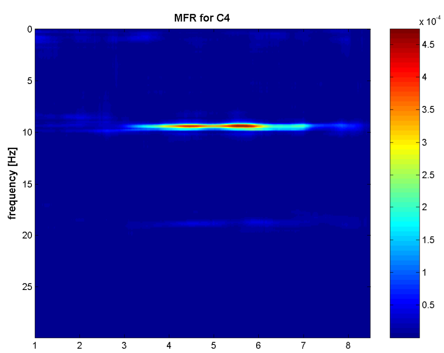
IV. RESULTS

A. *MFR Time-frequency mapping*

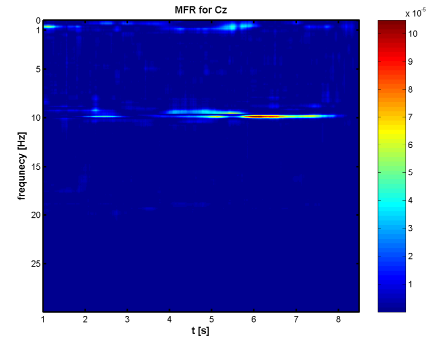
According to those described in Section III, MP algorithm combined with Fisher-Ratio were applied to processing EEG signals over C3, C4 and Cz, so that the corresponding MFR time-frequency distributive mapping of EEG spectral components are obtained and displayed in Figure 1.



(a)



(b)



(c)

Figure 1. MP-Fisher-Ratio (MFR) time-frequency distributive mapping of EEG spectral components

From Figs 1, it can be seen that the time-frequency mapping of Fisher-Ratio of EEG spectral components candidate feature sets within 8-30Hz and 0-9s are of characteristic as follows:

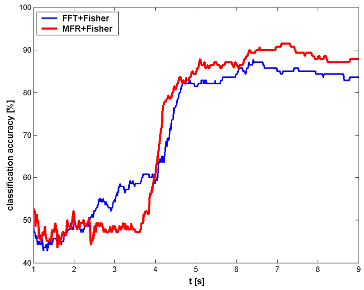
1) The most relevant EEG spectral frequencies corresponding to the bigger Fisher-Ratio are mainly concentrating on two narrow frequency bands revolved with μ and β rhythms respectively, which is consistent with the previous studies on ERD/ERS [2]. MFR time-frequency mapping of EEG spectral components provide the direct, instructive and useful pre-knowledge for properly selecting the most relevant EEG frequency components which have good seperability for left and right hand motor imagery tasks.

2) Although the electrodes C3 and C4 are located over two symmetric hemispheres, MFR time-frequency distribution is asymmetric. So the corresponding relevant EEG spectral components over the symmetric position C3 and C4 are not selected equally, which means that the parameters related to frequency bandwidth over C3, C4 should be chosen differently. The asymmetry of the relevant spectral components existing over C3 and C4 is consistent with those reported in [4].

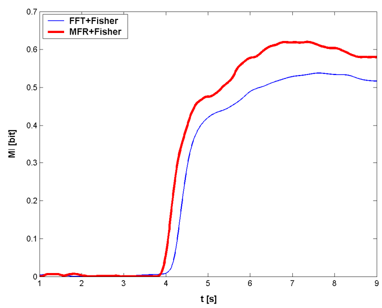
3) Currently, most of researchers emphasize EEG signals over C3 and C4 so that EEG signal over Cz is often neglected. From Fig.1, it can be clearly seen that the specific frequency components over Cz still have separability for two mental motor imagery tasks even though the degree of separability over Cz is not as big as that over C3 and C4. It can be expected that EEG spectral frequency components over Cz also may contain some useful information and help to improve classification performance. This finding is consistent with that of multichannel linear descriptors used for improving classification performance with features over Cz [10].

B. *Classification results*

Morlet wavelet is applied to extract the optimized EEG spectral components and FDA with information accumulation is applied to classify left and right hand motor imagery tasks offline. The classification results are showed in Fig 2. To verify the superiority of pre-selecting EEG spectral components with the proposed methods, the classification results without pre-selection of EEG spectral features within 8-30Hz over C3 C4 and Cz by FFT are also given.



(a)



(b)

Figure 2. Classification results; (a and b) display the classification accuracy and MI time course, in which the red and blue curves correspond to the results of using feature selection by MFR and non feature selection by FFT

From Fig 2, it can be clearly seen that Morlet wavelet filter combined with MFR time-frequency analysis for feature optimization obtain the satisfactory classification results, in which two indexes i.e. classification accuracy and MI are much better than that of non-pre-selecting EEG spectral components extracted by only FFT. To further quantify the efficiency of the optimized EEG features with the proposed methods in this paper, the values of two parameters including the maximum classification accuracy and maximum MI are given in TABLE 1, which gives the comparison of the quantitative classification results among these methods. To show the contribution of EEG feature information over Cz, the results with EEG feature over Cz and without EEG feature over Cz are also given and compared.

TABLE I. COMPARISON OF THE MAXIMUM CLASSIFICATION ACCURACY(%) AND MI (BIT) WITH DIFFERENT METHODS

	MFR+Morlet (C3,C4)	MFR+Morlet (C3,C4,Cz)	FFT(8-30Hz) (C3,C4,Cz)
Max Classification accuracy (%)	90.71	91.43	87.86
Max MI(bit)	0.6183	0.6199	0.5379

Obviously, for the maximum classification accuracy and MI, Morlet wavelet filter for extracting EEG features over C3 C4 and Cz based on MFR time-frequency analysis reaches the best results. Besides, Morlet wavelet filter for optimizing EEG spectral components result in the much better results than that of only FFT extracting the non-pre-selecting EEG features, which show that MFR time-frequency mapping can provide a useful and effective guide for the pre-selection of relevant EEG spectral components and verify the effectiveness of Morlet wavelet filter with high frequency resolution for improving the classification performance. Finally, adding the feature information from electrode Cz shows an improved classification results especially for MI, which indicates that EEG signal over Cz makes contribution to classification performance and should not be neglected.

V. CONCLUSTION

The experimental results show that pre-selection of the individual EEG spectral components could help to improve the accuracy of identifying the left and right hand motor imagery tasks. And the specific EEG spectral components over Cz also contributes to the classification of mental tasks and should be paid attention to. Finally, the results indicate that classification accuracy on optimized feature component is generally better than on all the frequency components. Only certain frequency bands carry all the relevant information and only these selected bands should be used online.

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